Cognitive Maps and Ghost Fences: Quantifying Behavioral State Dynamics Following Habitat Expansion in African Elephants

HACR

January 11, 2026

Abstract

Habitat fragmentation remains a primary threat to megaherbivore conservation, often addressed through physical fence removal. However, spatial access does not implicitly guarantee functional spatial use. We deployed a calibrated 4-state Hidden Markov Model (HMM) and integrated Step-Selection Analysis (iSSA) to examine the behavioral response of African elephants (*Loxodonta africana*) to fence removal at Kariega Game Reserve, South Africa. By classifying 30-minute GPS telemetry into Sleeping, Resting, Foraging, and Movement states, we quantified activity budgets and habitat selection across a Before-After-Control-Impact (BACI) experimental design. Our findings reveal a persistent "ghost fence" effect where cognitive memory of historical boundaries constrains current behavior. While physical crossings occurred almost immediately post-removal, they were dominated by transit-intensive movement states, with a significant delay in the establishment of foraging behaviors in novel zones. These results emphasize that conservation success hinges on behavioral adaptation rather than mere infrastructure removal.

1 Introduction

The restoration of landscape connectivity is a cornerstone of modern conservation strategy, particularly for wide-ranging megaherbivores like the African elephant (*Loxodonta africana*). Across Africa, the removal of boundary fences between private reserves and national parks has created vast, permeable landscapes intended to facilitate ancestral migration routes and mitigate human-wildlife conflict.

However, the efficacy of these corridors is mediated by the complex cognitive architecture of the species they serve. Elephants are characterized by long-term spatial memory, social learning, and a highly developed sense of risk perception. Their navigation is not merely a response to immediate environmental cues but is governed by internal cognitive maps that store decades of spatial information, including the location of historical barriers and associated threats.

When a physical barrier is removed, the transition from "accessible space" to "utilized habitat" is rarely instantaneous. A phenomenon known as the "ghost fence" effect may occur, where animals continue to exhibit avoidance or stress-associated behaviors at former boundary lines. Traditional movement analysis, which often relies on static occupancy points, fails to distinguish between an elephant that has successfully integrated a new area and one that is merely passing through it in a state of heightened vigilance.

To address this, we utilize a 4-state Hidden Markov Model (HMM) integrated with Resource Selection Functions (RSF). This dual framework allows us to partition continuous trajectories into functionally distinct behavioral states to predidict Sleeping, Resting, Foraging, and Movement, providing a high-resolution lens through which to examine behavioral plasticity and cognitive memory persistence in a cohort of elephants during a natural fence-removal experiment at Kariega Game Reserve.

2 Research Questions and Predictions

2.1 RQ1: Overall Activity Budgets

RQ1: How do African elephants allocate time among movement, foraging, sleeping, and low-energy behavioral states as inferred from GPS collar data?

- *Prediction 1.1:* Elephants will spend the largest proportion of time in low-energy and foraging states, with movement and sleeping comprising smaller fractions.
- *Prediction 1.2:* Activity budgets will vary seasonally, with increased movement during resource-limited periods.
- *Prediction 1.3:* Sleeping will be nocturnal and spatially clustered in low-disturbance areas.

2.2 RQ2: Spatial Distribution of Behavioral States

RQ2: How can patterns of state-specific selection be used to prioritize core habitats vs. corridors?

- *Prediction 2.1:* Foraging/low-energy states will be associated with core areas of high NDVI and resource richness.
- *Prediction 2.2:* Movement states will delineate functional corridors between core areas.

- *Prediction 2.3:* Sleeping states will highlight critical refugia characterized by cover and safety.

2.3 RQ3: Demographic and Social Variation

RQ3: How do budgets and selection differ by sex, age, and social herd tendencies?

- *Prediction 3.1:* Adult bulls will exhibit higher movement proportions and broader spatial use.
- *Prediction 3.2:* Family groups will allocate more time to foraging and remain closer to water/cover.
- *Prediction 3.3:* Exploration of novel habitats will be lead by specific socially exploratory herds.

2.4 RQ4: Persistence of "Ghost Fence" Effects

RQ4: Do elephants exhibit persistent constraints following fence removal?

- *Prediction 4.1:* Elephants will initially slow or "bounce" near former fence lines.
- *Prediction 4.2:* Ghost fence effects will decline over time as crossings increase.
- *Prediction 4.3:* Older or more conservative individuals will show longer persistence of memory.

3 Methodology

3.1 Data Acquisition and Preprocessing

High-resolution GPS telemetry data were collected from six African elephants (*Loxodonta africana*) across the Kariega Game Reserve and Harvestvale study areas. To facilitate robust behavioral segmentation and ensure analytical consistency across datasets with varying sampling rates, raw trajectories were cleaned and resampled to a uniform 30-minute temporal resolution using a five-minute tolerance window. All coordinate data were projected from WGS84 to UTM Zone 35S for accurate Euclidean distance calculations. To address limitations in likelihood estimation caused by true zero-displacement (GPS jitter), small random values between 0.0001 and 0.002 km were injected into stationary points, anchoring the "Sleeping" state as a distinct biological entity while preserving the continuity of the Gamma distributions.

3.2 Behavioral State Decoding via Hidden Markov Models

Movement trajectories were segmented into four discrete behavioral states using a multi-state Hidden Markov Model (HMM) fitted with Gamma distributions for step lengths and von Mises distributions for turning angles. The states were defined by four biological anchors derived from the study's *in-situ* observations: **Sleeping**, **Resting (Low-energy)**, **Foraging**, and **Movement** (refer to Table 1).

Unlike rigid thresholding, the HMM approach utilizes a probabilistic framework that accounts for the significant overlapping of movement distributions. For intermediate values where step lengths fall between state means, such as a 90m step occurring between the Foraging (60m) and Movement (250m) anchors, where the model resolves classifications using a hierarchical tie-breaking logic. The turning angle serves as the primary discriminator; directed steps (turning angles near 0°) are classified as Movement, while tortuous steps (turning angles near 180°) are attributed to search-based Foraging. Furthermore, the Viterbi algorithm was employed to maximize global behavioral consistency, ensuring that the inferred state at any given point is informed by the persistence of behaviors in the preceding and subsequent intervals. Diurnal and nocturnal transitions were incorporated as model covariates to further distinguish between deep nocturnal sleep cycles and daytime maintenance activities.

Table 1: Parameterization of the 4-State Hidden Markov Model (30-min Intervals)

State	**Mean Step (m)**	**SD Step (m)**	**Mean Angle**	**Conc. (κ)**	**Biological Driver**
Sleeping	1.2	1.0	180°	0.05	Nocturnal Inactivity
Resting	8.0	6.0	180°	0.20	Nurturing / Shade
Foraging	60.0	40.0	180°	0.40	Search / Browsing
Movement	250.0	120.0	0°	0.60	Directed Relocation

3.3 Experimental Design and Resource Selection

The study utilized a Before-After-Control-Impact (BACI) design centered on the removal of the Kariega West physical fence line. The temporal periods were segmented into three stages: **Pre** (physical constraint baseline), **Interim** (transitionary period), and **Post** (full territory adaptation). To model spatial preference, Resource Selection Functions (RSF) were integrated within an Integrated Step-Selection Analysis (iSSA) framework. For each observed location, five available locations were generated by sampling from the state-specific movement distributions identified by the HMM. Environmental selection was modeled at a 30m spatial resolution, incorporating MODIS-derived NDVI, land-cover classes, and distance-to-water as primary covariates.

3.4 Boundary Interactions and the Ghost Fence Effect

To quantify the "Ghost Fence" effect (cognitive memory persistence), spatial interaction analysis was conducted within discrete 50m, 100m, and 200m buffers of the former fence line. Interactions were categorized as **Crossings** or **Bounces**. A Bounce was algorithmically defined as an entry into the interaction buffer followed by a return to the zone of origin. The resolution of intermediate step lengths within these zones was critical for distinguishing between a "Resting" pause near the boundary and a "Foraging" search along the fence line (refer to Table 2).

Table 2: Resolution of Intermediate Step Lengths (Grey Areas)

Range (m)	**Classification**	**Primary Tie-Breaker**
1.5 – 8.0	Resting / Sleep	Temporal context (Night = Sleep)
8.0 – 60.0	Foraging / Rest	Tortuosity (High = Foraging)
60.0 – 120.0	Movement / Forage	Turning Angle ($< 45°$ = Movement)

3.5 Resource Selection Analysis (RSF)

To quantify habitat preference, we implemented Resource Selection Functions (RSF) within an integrated Step-Selection Analysis (iSSA) framework using the `amt` and `terra` packages in R. Selection was modeled by comparing observed GPS locations (*used*) against a set of *available* points (*pseudo-absences*) sampled from the study area's relevant availability masks (e.g., Kariega West for baseline periods and the expanded study area for post-removal periods). Background sampling was conducted at a 2:1 ratio relative to observed points to ensure a robust representation of environmental availability.

We extracted environmental values from a high-resolution 30m predictive stack comprising 41 layers, including MODIS-derived NDVI, distance-to-water, and terrain ruggedness (TRI). Selection intensity was estimated using Generalized Linear Models (GLMs) with a binomial logit link for each behavioral state and BACI period. Predicted probabilities were normalized to a 0–1 scale to generate high-resolution selection intensity maps. Model performance was validated using Area Under the ROC Curve (AUC) metrics, ensuring that the selection patterns accurately distinguished between utilized and available landscapes.

3.6 Interactive Web Visualization Platform

To facilitate multi-dimensional exploration of the behavioral and spatial datasets, we developed a custom interactive web visualization platform using a modern HTML5/JavaScript stack. The platform integrates `Leaflet.js` for high-performance geospatial rendering of trajectories and `Chart.js` for dynamic time-budget and temporal activity analysis. A key feature of the interface is the synchronous period comparison, allowing researchers to toggle between Pre-, Interim-, and Post-fence removal stages while maintaining consistent spatial perspectives. The platform serves as both a diagnostic tool for identifying behaviorally-mediated movement patterns and a decision-support system for visualizing predicted habitat suitability across different elephant cohorts.